MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **GENERAL ARRANGEMENT DRAWING REAGENTS- CAUSTIC SODA & CYANIDE EQUIP. GA & DETAILS OF GRADE SLAB**

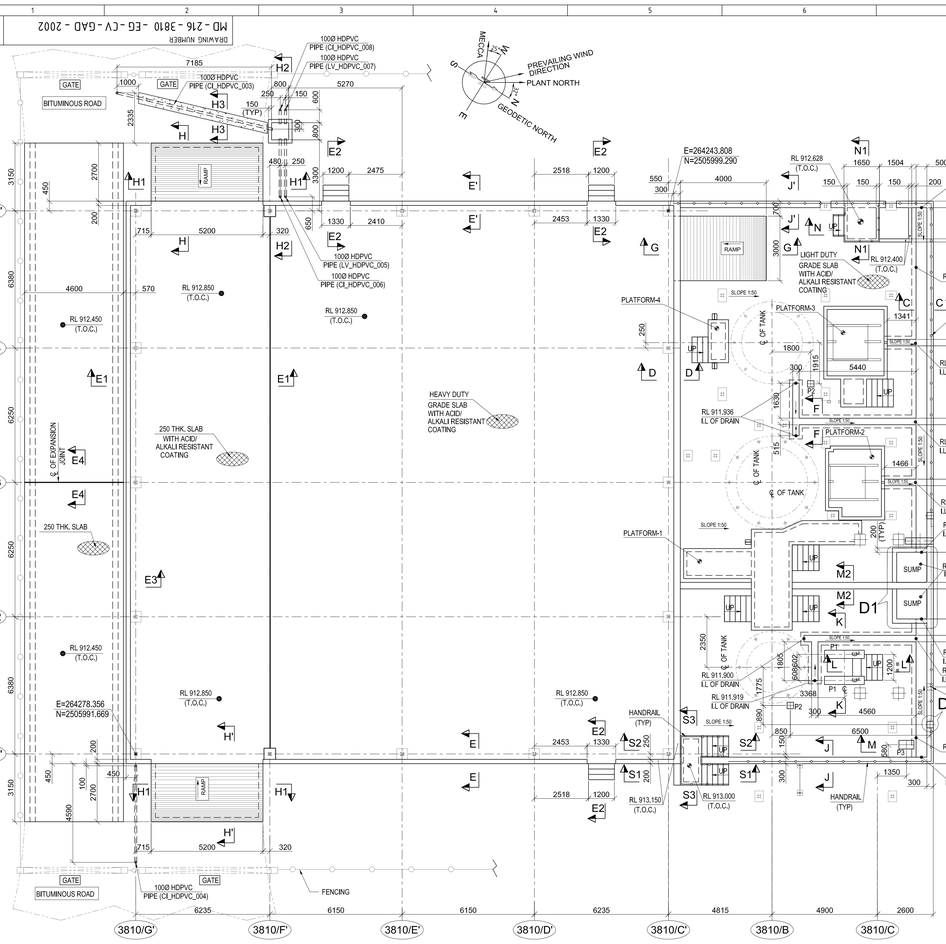
Document Number: **MD-216-3810-EG-CV-GAD-2002**
Revision: **D01**
Revision Date: **24.07.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AA24	MTL01	38128	OU100704669	8
L&T Document Number:	O19045	450	C	G	24496	001	8

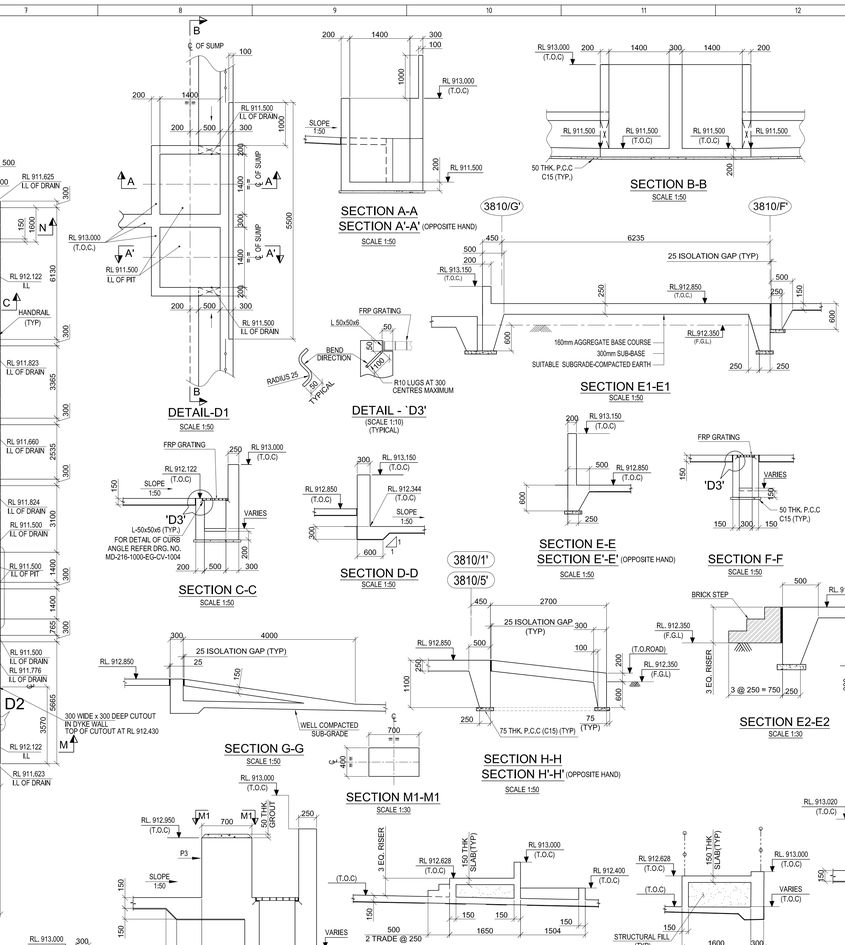
REVISION HISTORY

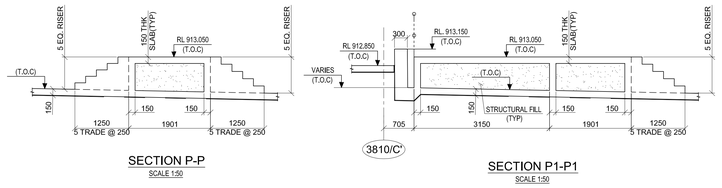
Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	20.10.2020	FIRST ISSUE	SOR	SJM	AG	
C01	22.12.2020	ISSUED FOR CONSTR. AND REVISED AS MARKED C01	SOR	SJM	AG	
C02	01.02.2021	ISSUED FOR CONSTR. AND REVISED AS MARKED C02	SOR	SJM	AG	
C03	22.02.2021	ISSUED FOR CONSTR. AND REVISED AS MARKED C03	SOR	SJM	AG	
C04	26.02.2021	ISSUED FOR CONSTR. AND REVISED AS MARKED C04	SOR	SJM	AG	
C05	25.03.2021	ISSUED FOR CONSTR. AND REVISED AS MARKED C05	SOR	SJM	AG	
C06	29.04.2021	ISSUED FOR CONSTR. AND REVISED AS MARKED C06	SOR	SJM	AG	
C07	16.06.2021	ISSUED FOR CONSTR. AND REVISED AS MARKED C07	SOR	SJM	AG	
D01	24.07.2023	AS BUILT	SOR	SJM	AG	

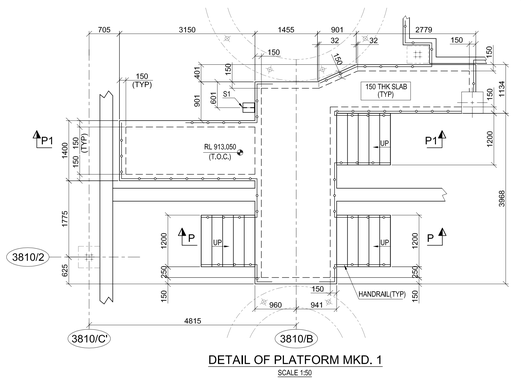
DRAWING NUMBER: MD-216-3810-EG-CV-GAD-2002 | REV: D01

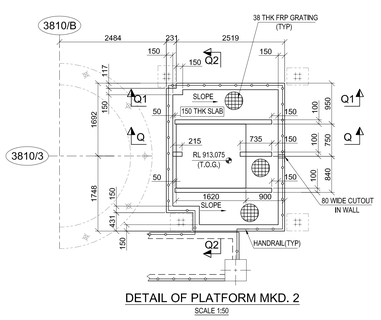


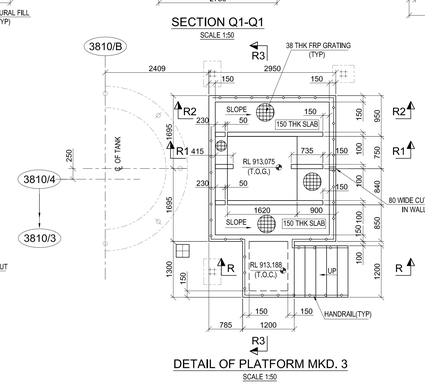
LAYOUT OF GRADE SLAB
(SCALE - 1:100)

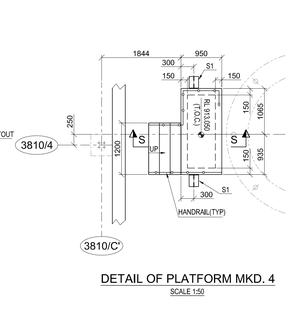


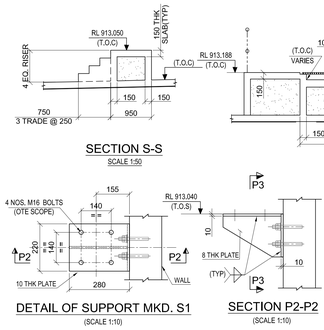












LEGEND :-
R.L - REDUCED LEVEL
I.L - INVERT LEVEL
THK. - THICK
TYP. - TYPICAL.
MKD. - MARKED
T.O.C - TOP OF CONCRETE
T.O.G - TOP OF GROUT
℄ - CENTRE LINE
P.C.C - PLAIN CEMENT CONCRETE
F.G.L. - FINISHED GROUND LEVEL.
GA - GENERAL ARRANGEMENT
EQ. - EQUAL
T.O.S - TOP OF STEEL
T.O.ROAD - TOP OF ROAD
Ø - DIAMETER
R - RADIUS

NOTES:-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 UNLESS NOTED OTHERWISE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420 MPa)
4. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE POCKETS ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
5. FOR REINFORCEMENTS DETAIL REF. DRG. NO. MD - 216 - 3810 - EG - CV - DED - 2002.
6. FOR STANDARD DETAILS FOR PAVING REF. DRG. NO : MD-216-1000-EG-CV-DED-1002.
7. FOR FOUNDATION BOLT DETAILS REF. DRG. NO : MD-216-1000-EG-CV-DED-1003.
8. FOR GENERAL NOTE FOR CIVIL WORK REF. DRG. NO : MD-216-1000-EG-CV-DED-1001.
9. CHEMICAL RESISTANT COATING SHALL BE APPLIED TO THE ALL EXPOSED SURFACE OF CONCRETE INSIDE BUND WALL TO PROTECT AGAINST ACID / ALKALI.

SHEET 1 OF 2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	20/7/2023	AS BUILT	PML	SJM	SOR	AG	
		C07	15/06/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	PML	SJM	SOR	AG	
		C06	29/04/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	PML	SJM	SOR	AG	
		C05	25/03/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	PML	SJM	SOR	AG	
		C04	26/02/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	PML	SJM	SOR	AG	
	C07 15/06/2021	C03	22/02/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	PML	SJM	SOR	AG	
		C02	01/02/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	PML	SJM	SOR	AG	
MD-216-3810-EG-ME-GAD-1011	REAGENTS - CAUSTIC AND CYANIDE	C01	22/12/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	PML	SJM	SOR	AG	
	PLAN VIEW	A01	20/10/2020	ISSUED FOR REFERENCE	PML	SJM	SOR	AG	

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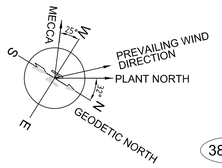
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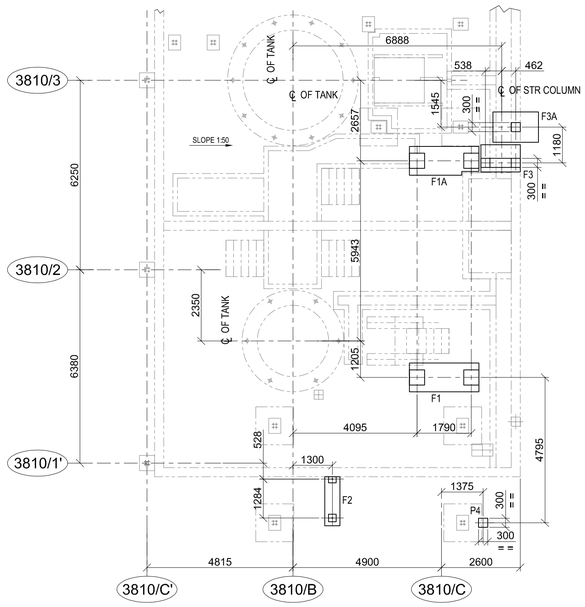
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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24496_001 | REVISION: 8
OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ AA24_ MTL01 _ 38128 | DOCUMENT ID: OU100704669 | REVISION: 8

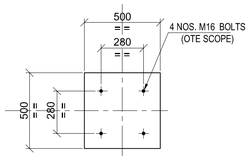
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CHECKED	SJM	16/10/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	SOR	14/10/2020	REAGENTS - CAUSTIC SODA & CYANIDE EQUIPMENT
APPROVED	AG	20/10/2020	GA & DETAILS OF GRADE SLAB
EMPLOYER			
SCALE: 1:100, 1:50, 1:25, 1:20		A0	DRAWING NUMBER: MD - 216 - 3810 - EG - CV - GAD - 2002 REV: D01

DRAWING NUMBER MD-216-3810-EG-CV-GAD-2002 REV D01

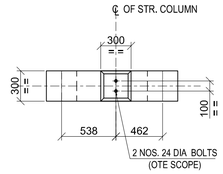




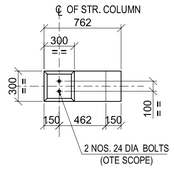
LAYOUT OF FOUNDATION
(SCALE - 1:100)



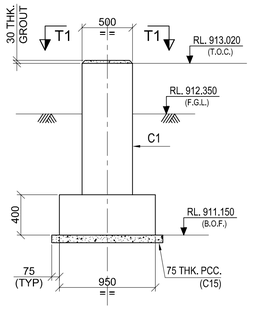
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SCALE 1:20



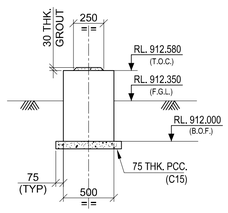
SECTION : V1-V1
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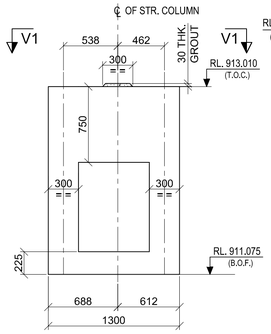
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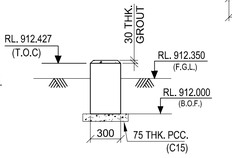
SECTION : T-T
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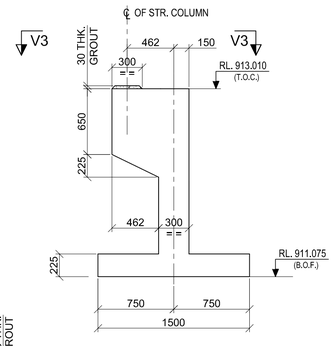
SECTION U-U
SCALE 1:30



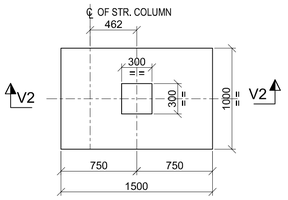
SECTION V-V
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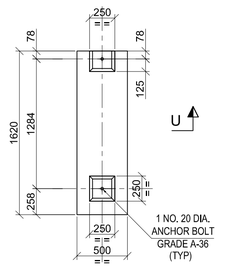
DETAIL OF P4
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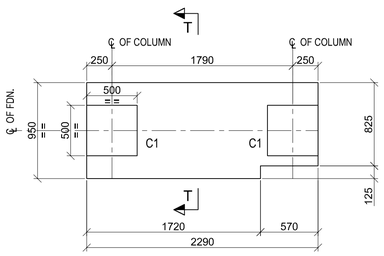
SECTION V2-V2
SCALE 1:30



DETAILS OF FOUNDATION MKD. F3A
(1 NO REQD.)
SCALE 1:30

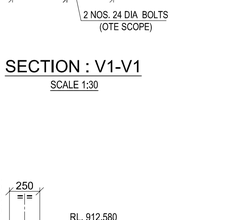


DETAIL OF FOUNDATION MKD F2
(1 NO REQUIRED)
SCALE 1:30



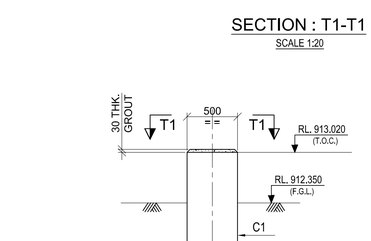
DETAILS OF FOUNDATION MKD. F1A
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SCALE 1:30

SECTION H3-H3
SCALE 1:25



DETAIL-D4
SCALE 1:25

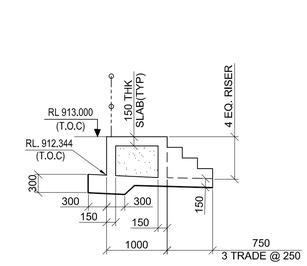
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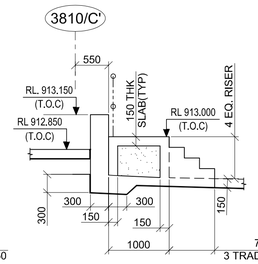
DETAILS OF FOUNDATION MKD. F1
(1 NO. REQD.)
SCALE 1:30

TYPICAL DETAIL OF CONDUIT
SCALE 1:50

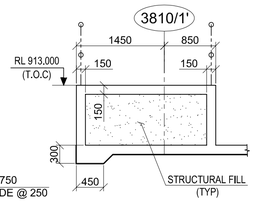
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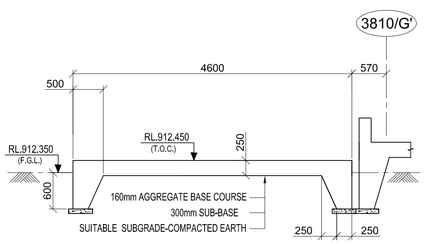
SECTION S1-S1
SCALE 1:50



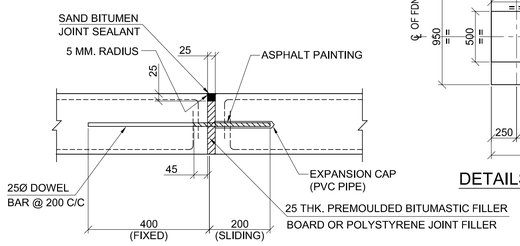
SECTION S2-S2
SCALE 1:50



SECTION S3-S3
SCALE 1:50



SECTION E3-E3
SCALE 1:50



SECTION E4-E4
(SCALE 1:10)

LEGEND :-

R.L. - REDUCED LEVEL
I.L. - INVERT LEVEL
THK. - THICK
TYP. - TYPICAL.
MKD. - MARKED
T.O.C - TOP OF CONCRETE
T.O.G - TOP OF GROUT
℄ - CENTRE LINE
P.C.C - PLAIN CEMENT CONCRETE
F.G.L. - FINISHED GROUND LEVEL
GA - GENERAL ARRANGEMENT
B.O.F. - BOTTOM OF FOUNDATION
DIA - DIAMETER
CJ - CONSTRUCTION JOINT
FDN. - FOUNDATION
STR. - STRUCTURAL
EQ. - EQUAL
REQD. - REQUIRED
Ø - DIAMETER
R - RADIUS
RCC - REINFORCED CEMENT CONCRETE
CHQD - CHEQUERED

SCHEDULE OF CONDUIT :-

CONDUIT MKD.	ELEVATION AT ℄ OF CONDUIT
CI_HDPVC_003	RL 911.650
CI_HDPVC_004	RL 911.650
LV_HDPVC_005	RL 911.650
CI_HDPVC_006	RL 911.650
LV_HDPVC_007	RL 911.650
CI_HDPVC_008	RL 911.650

NOTES:-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) FOR FOUNDATION & COLUMN.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 350KN/SQ.M.
5. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
6. FOR STANDARD REINFORCEMENT DETAIL & GENERAL NOTES FOR CIVIL WORKS REFER DRAWING NO, MD-216-1000-EG-CV-DED-1005 &1001.
7. FOR REINFORCEMENTS DETAIL REF. DRG. NO. MD - 216 - 3810 - EG - CV - DED - 2002.

SHEET 2 OF 2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	21/1/2023	AS BUILT	PML	SJM	SOR	AG	
		C07	15/06/2021	ISSUED FOR CONSTRUCTION	PML	SJM	SOR	AG	
		C06	29/04/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED △C06	PML	SJM	SOR	AG	
		C05	25/03/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED △C05	PML	SJM	SOR	AG	
		C04	26/02/2021	ISSUED FOR CONSTRUCTION	PML	SJM	SOR	AG	
		C03	22/02/2021	ISSUED FOR CONSTRUCTION	PML	SJM	SOR	AG	
		C02	01/02/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED △C02	PML	SJM	SOR	AG	
MD-216-3810-EG-ME-GAD-1011	REAGENTS - CAUSTIC AND CYANIDE	C01	22/12/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED △C01	PML	SJM	SOR	AG	
	PLAN VIEW	A01	20/10/2020	ISSUED FOR REFERENCE	PML	SJM	SOR	AG	

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Saudi Arabian Mining Company شركة التعدين العربية السعودية
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DRAWN	PML	14/10/2020
CHECKED	SJM	16/10/2020
ENGINEER	SOR	14/10/2020
APPROVED	AG	20/10/2020
EMPLOYER		
SCALE 1:30		A1

LARSEN & TOUBRO DRAWING NUMBER 019045_450_C_G_24496_001 — REVISION 8
OUTOTEC DRAWING NUMBER 900707__GAU01__AA24_MTL01__38128 — DOCUMENT ID OU100704669 — REVISION 8

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
REAGENTS - CAUSTIC SODA & CYANIDE EQUIPMENT
GA & DETAILS OF GRADE SLAB

DRAWING NUMBER MD-216-3810-EG-CV-GAD-2002 — REV D01